

KYRGYZSTAN

Country Risk Profile · Central Asia · 45 events on record · 1911–2025

78

Hazard Exposure

High seismic frequency + annual flood/landslide events

80

Vulnerability

Informal settlements, 50%+ worn water network (Osh)

82

Future Risk (2050)

Glacier melt, water deficit, temperature rise

50

Data Confidence

Partial coverage — 16 of 45 events have quantitative data

Executive Summary

Kyrgyzstan's risk landscape is defined by the combination of active seismicity, dense informal settlements on hazard-prone slopes, and rapid glacier-driven hydrological change. The record extends to 1911 (Chon-Kemin earthquake, 452 deaths), giving one of the longest time horizons in the regional dataset. The 2008 Nura earthquake (75 deaths) and 2003–2004 landslide cluster (67 deaths across two events) illustrate the persistent geophysical threat. Osh city — 478,000 population, growing at 8% annually on seismic slopes — represents the single highest urban concentration risk in the region. Projected 30% water deficit by 2030 from glacier retreat will compound food and livelihood insecurity significantly.

RECORDED DISASTER IMPACTS

2,927

Recorded deaths

16 / 45 events — dominated by 1911 EQ (452)

259K

People affected (known)

12 / 45 events with data

\$5M

Recorded economic losses

3 / 45 events — extreme undercount

HAZARD PROFILE

Hazard Group	Events	Share
Hydrological	24	53%
Geophysical	13	29%
Biological	4	9%
Meteorological	4	9%

Dominant hazard: Floods & Flash Floods (53%). Geophysical (Earthquakes, Landslides, Mudslides) account for 29% but historically the deadliest. No volcanic or geothermal risk.

⚠ Figures are partial sums from events with available data. Most events lack quantitative records. Actual losses are substantially higher.

Selected Significant Events

Year	Event	Deaths	Affected	Econ. Loss	Location
2025	Epidemic	2	8,211	—	Bishkek, Chuy, Jalal-Abad, Osh
2024	Flood / Mudslide	5	720	—	Osh City, Jalal-Abad

2021	Mudslide + Flood	9	—	—	Aksy & Suzak, Jalal-Abad
2020	COVID-19	2614	179K	—	Kyrgyzstan (national)
2008	Earthquake	75	150	—	Nura, Kura villages
2003–04	Landslides (×2)	67	12	—	Sogot / Alay, Osh region
1992	Earthquake	75	—	—	Suusamyr
1911	Earthquake	452	740	—	Chon-Kemin area

Key Risk Indicators

Domain	Indicator	Value	Confidence	Source / Note
Disaster Record	Recorded deaths (partial)	2,927	Low	16 of 45 events; 1911 Chon-Kemin EQ dominates
Disaster Record	Deadliest recent event	75 dead	High	2008 Earthquake, Nura/Kura villages
Disaster Record	Landslide cluster 2003–2006	41 dead	Medium	3 events, Osh / Jalal-Abad
Infrastructure	Water network decay (Osh)	50%+	Low	Osh Risk Profile 2025 — subject to verification
Infrastructure	Population density (Osh ctr)	9,000/km ²	High	Osh City Administration 2025
Infrastructure	Osh annual urban growth	8%/yr	High	Osh City Administration 2025
Climate & Future	Projected water deficit 2030	–30%	Medium	UNDP NAP Kyrgyzstan
Climate & Future	Temperature rise by 2090	+6.2°C	Low	SSP5-8.5 high-emission scenario
Climate & Future	Papan dam breach scenario	113K m ³ /s	Low	Peak outflow estimate, Osh Risk Profile 2025
Economic Exposure	Recorded economic losses	\$5M	Low	3 of 45 events — severe undercount
Economic Exposure	Agriculture water dependency	High	Medium	Glacier melt threatens irrigation supply

Policy & Commitment Tracker (NAP / NDC)

Area	Indicator	Direction	Current / Target	Source
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Disaster Risk	National DRR Strategy 2025–2030	↗ Right Direction	Adopted → Sendai aligned	Kyrgyzstan DRR Strategy 2025
Disaster Risk	Early warning (floods & landslides)	→ Off Track	Partial → Full coverage	UNDP Kyrgyzstan DRR 2024
Disaster Risk	Osh urban risk reduction plan	→ Off Track	Draft → Adopted by 2025	Osh City Master Plan 2025
Climate	NDC target vs 2015 baseline	↗ Right Direction, No Target	44% → 44% by 2030	Kyrgyzstan NDC 2021
Climate	Climate finance mobilised	? Insufficient Data	— → \$1.5B needed	NDC 2021
Climate	Glacier monitoring programme	↗ Right Direction	Active → Maintain	UNDP GEF Glacier Project 2023
Adaptation	Flood/landslide risk mapping	↗ Right Direction	60% → 100% by 2027	Kyrgyzstan NAP 2022
Adaptation	Informal settlement risk reduction (Osh)	↘ Wrong Direction	Stalled → Phase 1 by 2024	UNDP Osh Risk Profile 2025

Country Context — World Bank Indicators

7.1M Population	\$5,700 GDP / capita (PPP)	33.3% Below poverty line	81.5% Safe water access	37.8% Urban population
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Source: World Bank Open Data. Most recent available values (2022–2024).

Data Notes & Sources

Risk layer scores (Hazard Exposure, Vulnerability, Future Risk) are indicative composites derived from available data. They are not standardised national statistics and should not be used for inter-country ranking without methodological alignment. All disaster figures are partial — most events lack complete quantitative records. Economic loss figures in particular represent severe undercounts. Policy trend assessments are qualitative judgements based on available documents and should be validated by country office teams before external sharing.

- Kyrgyzstan Disaster Risk Profile — UNDP Istanbul Regional Hub, 2025
- Osh City Risk Profile — UNDP Urban Resilience Project, 2025
- Kyrgyzstan NDC (Nationally Determined Contribution) — 2021
- Kyrgyzstan NAP (National Adaptation Plan) — 2022
- Kyrgyzstan DRR Strategy 2025–2030
- UNDP GEF Glacier Monitoring Project — 2023
- World Bank Open Data — 2023/2024
- DesInventar / EM-DAT / IFRC / USGS / OCHA disaster event records
- UNDP Dushanbe GCAP — 2022 (regional reference)

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